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RESEARCH Network and Information Security

INTERESTS Decentralized Systems
Trusted and Privacy-preserving Computing
Mobile Communications and Network Security
Cyber-Physical Security

PROFESSIONAL **Assistant Professor** 08/2022 – Present

APPOINTMENT Department of Computer Science, University of Kentucky, KY, USA

Graduate Research Assistant 08/2017 – 05/2022

Department of Electrical and Computer Engineering, Virginia Tech, VA, USA

Intern 05/2016 – 08/2016

Mathematics and Modeling Department, Schlumberger-Doll Research, MA, USA

EDUCATION **Virginia Polytechnic Institute and State University**, VA, USA 08/2017 – 05/2022

– Ph.D. in Computer Engineering

University of Michigan, MI, USA 09/2015 – 04/2017

– M.S. in Electrical Engineering-Systems

Shanghai Jiao Tong University, Shanghai, China 09/2010 – 06/2014

– B.S.E. in Information Engineering

– (Secondary) B.Econ. in Finance

CONFERENCE (underlined: student directly supervised; *: I am the corresponding author.)

PUBLICATIONS

[1] Model-Consistent Byzantine-Resilient Decentralized Federated Learning for Collaborative Missions

Y. Li, S. Bhujel, C. Lira, N. Wang, **Y. Xiao**

IEEE Conference on Communications and Network Security 2026 (**CNS'26**), September 2026.

[2] SpexPay: A Privacy-Preserving Pay-As-You-Go System for Dynamic Spectrum Sharing

M. M. A. Barat, H. Yu, S. Li, **Y. Xiao**, Y. Shi, E. Burger, Y. T. Hou, W. Lou

European Symposium on Research in Computer Security 2026 (**ESORICS'26**), September 2026.

[3] zkSAS: Practical Zero-knowledge Proofs for Verifiable Spectrum Access Management

N. F. Purbasha, I. Alom, E. Burger, Y. T. Hou, W. Lou, **Y. Xiao**

IEEE International Symposium on Spectrum Innovation 2026 (**DySPAN'26**), May 2026.

[4] IU-GUARD: Privacy-Preserving Spectrum Coordination for Incumbent Users under Dynamic Spectrum Sharing

S. Li, H. Yu, S. Shi, M. M. A. Barat, **Y. Xiao**, Y. T. Hou, W. Lou

IEEE International Conference on Communications 2026 (**ICC'26**), May 2026.

[5] FC-GUARD: A Privacy-Preserving Exchange System Between Fiat Currency and Cryptocurrency

S. Li, H. Yu, M. M. A. Barat, **Y. Xiao**, Y. T. Hou, W. Lou

IEEE Conference on Computer Communications 2026 (**INFOCOM'26**), May 2026.

[6] AnonyCall: Enabling Native Private Calling in Mobile Networks

H. Yu, C. Zhang, **Y. Xiao**, A. Keromytis, Y. T. Hou, W. Lou

- Network and Distributed System Security Symposium 2024 (**NDSS'26**), February 2026.
- [7] Closing the Visibility Gap: A Monitoring Framework for Verifiable Open RAN Operations
H. Yu, M. M. Al Barat, **Y. Xiao**, Y. T. Hou, W. Lou
IEEE Conference on Communications and Network Security 2025 (**CNS'25**), September 2025.
 - [8] Scale-MIA: A Scalable Model Inversion Attack against Secure Federated Learning via Latent Space Reconstruction
S. Shi, N. Wang, **Y. Xiao**, C. Zhang, Y. Shi, Y. T. Hou, W. Lou
Network and Distributed System Security Symposium 2025 (**NDSS'25**), February 2025.
 - [9] TriSAS: Toward Dependable Inter-SAS Coordination with Auditability
S. Shi, **Y. Xiao**, C. Du, Y. Shi, C. Wang, R. Gazda, Y. T. Hou, E. Burger, L. DaSilva, W. Lou
19th ACM ASIA Conference on Computer and Communications Security (**ASIACCS'24**), July 2024.
 - [10] AAKA: An Anti-Tracking Cellular Authentication Scheme Leveraging Anonymous Credentials
H. Yu, C. Du, **Y. Xiao**, A. Keromytis, C. Wang, R. Gazda, Y. T. Hou, W. Lou
Network and Distributed System Security Symposium 2024 (**NDSS'24**), February 2024.
 - [11] Rethinking Single Sign-On: A Reliable and Privacy-Preserving Alternative with Verifiable Credentials
A. D. Johnson, I. Alom, **Y. Xiao***
10th ACM Workshop on Moving Target Defense (**MTD**), November 2023. (Collocated with CCS 2023)
 - [12] Bijack: Breaking Bitcoin Network with TCP Vulnerabilities
S. Li, S. Shi, **Y. Xiao**, C. Zhang, Y. T. Hou, W. Lou
28th European Symposium on Research in Computer Security (**ESORICS'23**), September 2023.
 - [13] UCBlocker: Unwanted Call Blocking Using Anonymous Authentication
C. Du, H. Yu, **Y. Xiao**, Y. T. Hou, A. Keromytis, W. Lou
32nd USENIX Security Symposium (**USENIX Security'23**), August 2023.
 - [14] ARI: Attestation of Real-time Mission Execution Integrity
J. Wang, Y. Wang, A. Li, **Y. Xiao**, R. Zhang, W. Lou, Y. Hou, N. Zhang
USENIX Security Symposium (**USENIX Security'23**), August 2023.
 - [15] MS-PTP: Protecting Network Timing from Byzantine Attacks
S. Shi, **Y. Xiao**, C. Du, M. Shahriar, A. Li, N. Zhang, Y. Hou, W. Lou
16th ACM Conference on Security and Privacy in Wireless and Mobile Networks (**WiSec'23**), May 2023.
 - [16] A Decentralized Truth Discovery Approach to the Blockchain Oracle Problem
Y. Xiao, N. Zhang, W. Lou, Y. T. Hou
IEEE Conference on Computer Communications 2023 (**INFOCOM'23**), May 2023.
 - [17] Squeezing More Utility via Adaptive Clipping on Differentially Private Gradients in Federated Meta-Learning
N. Wang, **Y. Xiao**, Y. Chen, N. Zhang, W. Lou, Y. T. Hou
38th Annual Computer Security Applications Conference (**ACSAC'22**), December 2022.
 - [18] Mobile Tracking in 5G and Beyond Networks: Problems, Challenges, and New Directions
C. Du, H. Yu, **Y. Xiao**, W. Lou, C. Wang, R. Gazda, Y. T. Hou
19th IEEE International Conference on Mobile Ad-Hoc and Smart Systems (**MASS'22**), October 2022.
 - [19] CANShield: Signal-based Intrusion Detection for Controller Area Networks
M. H. Shahriar, **Y. Xiao**, P. Moriano, W. Lou, Y. T. Hou
9th Embedded Security in Cars USA conference (**escar USA'22**), June 2022.
 - [20] FLARE: Defending Federated Learning against Model Poisoning Attacks via Latent Space Representations
N. Wang, **Y. Xiao**, Y. Chen, Y. Hu, W. Lou, Y. T. Hou
17th ACM ASIA Conference on Computer and Communications Security (**ASIACCS'22**), May 2022.

- [21] Session Key Distribution Made Practical for CAN and CAN-FD Message Authentication
Y. Xiao, S. Shi, N. Zhang, W. Lou, Y. T. Hou
36th Annual Computer Security Applications Conference (ACSAC'20), December 2020.
- [22] PrivacyGuard: Enforcing Private Data Usage Control with Blockchain and Off-chain Contract Execution
Y. Xiao, N. Zhang, J. Li, W. Lou, Y. T. Hou
25th European Symposium on Research in Computer Security (ESORICS'20), September 2020.
- [23] Modeling the Impact of Network Connectivity on Consensus Security of Proof-of-Work Blockchain
Y. Xiao, N. Zhang, W. Lou, Y. T. Hou
IEEE International Conference on Computer Communications 2020 (INFOCOM'20), July 2020.

BOOK
CHAPTERS

- [1] Basic Proof-of-Stake Consensus Mechanisms
Y. Xiao*, W. Sun
Proof-of-Stake for Blockchain Networks: Fundamentals, challenges and approaches, IET, 2024
- [2] Distributed Consensus Protocols and Algorithms
Y. Xiao, N. Zhang, J. Li, W. Lou, Y. T. Hou
Blockchain for Distributed Systems Security, Wiley & Sons, 2019

JOURNAL
PUBLICATIONS

- (underlined: student directly supervised; *: I am the corresponding author.)
- [1] DEXO: A Verifiable and Scalable Oracle Mechanism for Inter-DApp Data Exchange.
Y. Li, I. Alom, W. Sun, Y. Xiao*
In *IEEE Internet of Things Journal (IOT-J)*, June 2025.
 - [2] CANShield: Deep Learning-Based Intrusion Detection Framework for Controller Area Networks at the Signal-Level
M. H. Shahriar, Y. Xiao, P. Moriano, W. Lou, Y. T. Hou
In *IEEE Internet of Things Journal (IOT-J)*, December 2023.
 - [3] BD-SAS: Enabling Dynamic Spectrum Sharing in Low-trust Environment
Y. Xiao, S. Shi, W. Lou, C. Wang, X. Li, N. Zhang, Y. T. Hou, J. H. Reed
In *IEEE Transactions on Cognitive Communications and Networking (TCCN)*, 2023.
 - [4] SofitMix: A Secure Offchain-Supported Bitcoin-Compatible Mixing Protocol
H. Xie, S. Fei, Z. Yan, Y. Xiao
In *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2022.
 - [5] MANDA: On Adversarial Example Detection for Network Intrusion Detection System
N. Wang, Y. Chen, Y. Xiao, Y. Hu, W. Lou, Y. T. Hou
In *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2022.
 - [6] Decentralized Spectrum Access System: Vision, Challenges, and a Blockchain Solution
Y. Xiao, S. Shi, W. Lou, C. Wang, X. Li, N. Zhang, Y. T. Hou, J. H. Reed
In *IEEE Wireless Communications*, 2022.
 - [7] Challenges and New Directions in Securing Spectrum Access Systems
S. Shi, Y. Xiao, W. Lou, C. Wang, X. Li, Y. T. Hou, J. H. Reed
In *IEEE Internet of Things Journal (IOT-J)*, 2021.
 - [8] A Survey of Distributed Consensus Protocols for Blockchain Networks
Y. Xiao, N. Zhang, W. Lou, Y. T. Hou
In *IEEE Communications Surveys & Tutorials (COMST)*, 2020.
 - [9] Offloading Decision in Edge Computing for Continuous Applications Under Uncertainty
W. Chang, Y. Xiao, W. Lou, G. Shou

In *IEEE Transactions on Wireless Communications (TWC)*, 2020.

[10] Performance Analysis of Random Access Network with Post-backoff

C. Bu, **Y. Xiao**, T. Ye, P. Wu, X. Zhang, J. Wu

In *Telecommunications Science*, 2016.

TALKS, INTERVIEWS	Distributed Consensus Protocols for Blockchain – Invited virtual talk, <i>IEMS 5725 Blockchain and Applications</i> , CUHK On Mobile Network Security – Invited talk on mobile tracking and authentication, UK CyberCon’24 – TV interviews on the anti-tracking mobile network project, UK Now and LEX 18 – TV interview on mobile apps security for online sports betting, Fox 56 Blockchain and Trusted Execution Environment: Security Properties, Synergies, Applications – Invited talk, <i>Delta 2022 Cybersecurity Conference, UTN Facultad Regional Delta, Argentina</i> A Layered View towards Blockchain: From Consensus Security to Smart Contract Application – Invited talk, <i>CS Graduate Seminars</i> , Virginia Tech	4/2024 10/2024 12/2023 10/2023 11/2022 11/2020
WORKSHOPS, POSTERS	Session Key Distribution Made Practical for CAN and CAN-FD Message Authentication – Workshop talk, <i>Learning from Authoritative Security Experiment Results (LASER) Workshop</i> Enforcing Private Data Usage Control with Blockchain and Attested Off-chain Contract Execution – Poster, 2019 SaTC PI Meeting, NSF, Alexandria, VA A Graph Random-Walk Based Sampling Algorithm for Load Balancing in Cloud Systems – Poster, <i>6th Midwest Workshop on Control and Game Theory</i> , Ann Arbor, MI	12/2020 04/2019 04/2017
GRANTS	US National Science Foundation – “CAREER: Foundations of Operational Resilience and Secure Communication for Networked Real-Time Systems,” NSF Award #2442382, 07/15/2025–06/30/2030, \$534,264 (PI, sole) – “Collaborative Research: A Novel Architecture for Secure and Adaptive Spectrum Management Enabling Registration-Free Federal Users,” NSF Award #2433905, 09/01/2025–08/31/2028, \$237,500 (PI, non-lead) – “Collaborative Research: SaTC: CORE: Medium: An Anti-tracking and Robocall-free Architecture for Next-G Mobile Networks,” NSF Award #2247561, 05/01/2023–04/30/2027, \$300,000 (PI, non-lead). REU supplement in 2025: \$10,240. Office of Naval Research – “Byzantine Resilient Federated Learning in Sporadically Connected Wireless Networks, ONR Award #N00014-24-1-2730, subawarded through Virginia Tech, 10/01/2024–09/30/2027, \$300,000 (Co-PI)	
TEACHING	Instructor , University of Kentucky – CS 270: Systems Programming – CS 371: Introduction to Computer Networking – CS 378: Introduction to Cryptology – CS 572: Network Security – CS 585/685: Intermediate/Special Topics in CS: Blockchain Technologies Guest Lecturer , Virginia Tech – CS 5560: Fundamentals of Information Security - Cryptocurrency & Blockchain Graduate Teaching Assistant , Virginia Tech – ECE 2534: Microcontroller Programming and Interfacing	Fall 2025, 2026 Spring 2023, 2024 Spring 2025, 2026 Fall 2023, 2024 Fall 2022 Spring 2019 Fall 2017
ADVISING	PhD Students	

- Yue Li 08/2023 - Now
- Ifteher Alom 08/2023 - Now
- Sudip Bhujel 08/2024 - Now
- Nishat F. Purbasha 08/2025 - Now
- Mehedi H. Munna 05/2026 - Now

Master Students

- Yue Li 12/2022 - 05/2023
- Maxwell Meier 12/2025 - 05/2026

Undergrad Students

(EURF: Engineering Undergraduate Research Fellow; REU: NSF Research Experiences for Undergraduates)

- Cameron Lira (EURF) 05/2025 - 08/2023
- John Hostettler (REU) 05/2025 - 07/2023
- Gavin Prewitt (REU) 05/2025 - 07/2023
- Athan Johnson (EURF) 05/2023 - 09/2023
- William Rillo (Lab Assistant) 01/2025 - 05/2025

PhD Advisory Committee

- Xu Tao 04/2023 – 04/2025
- Yuhang Jiang 03/2023 – 12/2025

Master Advisory Committee

- Jacob Sobota 03/2023 – 04/2023
- Franklin Stokan 02/2023 – 04/2023
- Samuel Armstrong 10/2022 – 03/2023

ACADEMIC SERVICES

Conference Organization

- IEEE International Conference on Computer Communications (INFOCOM) 2023, 2024: Web Chair
- IEEE Conference on Communications and Network Security (CNS) 2020: Web Chair

Technical Program Committee

- IEEE Symposium on Security and Privacy (SP) 2027
- USENIX Security Symposium 2027
- Network and Distributed System Security (NDSS) Symposium 2026, 2027
- Annual Computer Security Applications Conference (ACSAC) 2024–2026
- IEEE International Conference on Computer Communications (INFOCOM) 2024–2026
- IEEE Conference on Communications and Network Security (CNS) 2026
- IEEE Military Communications Conference (MILCOM) 2023–2026
- International Conference on Computer Communications and Networks (ICCCN) 2023

Panelist

- NSF Panelist 2023
- NSF SaTC PI Meeting Undergraduate Track - Student Panelist 2019

Journal Review

- IEEE/ACM Transactions on Networking (TON)
- IEEE Internet of Things Journal (IOT-J)
- IEEE Transactions on Dependable and Secure Computing (TDSC)
- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Transactions on Communications (TCOM)
- IEEE Transactions on Computers (TC)
- ACM Transactions on Privacy and Security (TOPS)
- International Journal of Intelligent Systems (INT2)
- ACM Transactions on Cyber-Physical Systems (TCPS)

UNIVERSITY	Department of Computer Science, University of Kentucky	
SERVICES	– Committee on Higher Degrees – Hiring Committee – Faculty Member, Cybersecurity Certificate Program – Co-director, Cybersecurity Certificate Program	09/2022 - 08/2024 09/2024 - Now 04/2023 - 06/2026 07/2026 - Now
AWARDS	Distinguished Member of the INFOCOM 2024 Technical Program Committee – Awarded by the INFOCOM 2024 organizing committee INFOCOM 2020 Student Travel Grant – Awarded by the INFOCOM 2020 organizing committee BitShares Graduate Fellowship – Awarded by Virginia Tech CS Department, funded by BitShares Inc. Completion of the Elite Engineer Cultivation Program in Information Engineering – Certified by Shanghai Jiao Tong University First Prize in the 28th National Physics Contest of College Students (Shanghai Division) – Awarded by Shanghai Physics Society	2024 2020 2019 2014 2011
PROFESSIONAL MEMBERSHIPS	– IEEE Member, Communications Society, Computer Society	11/2017 – Present